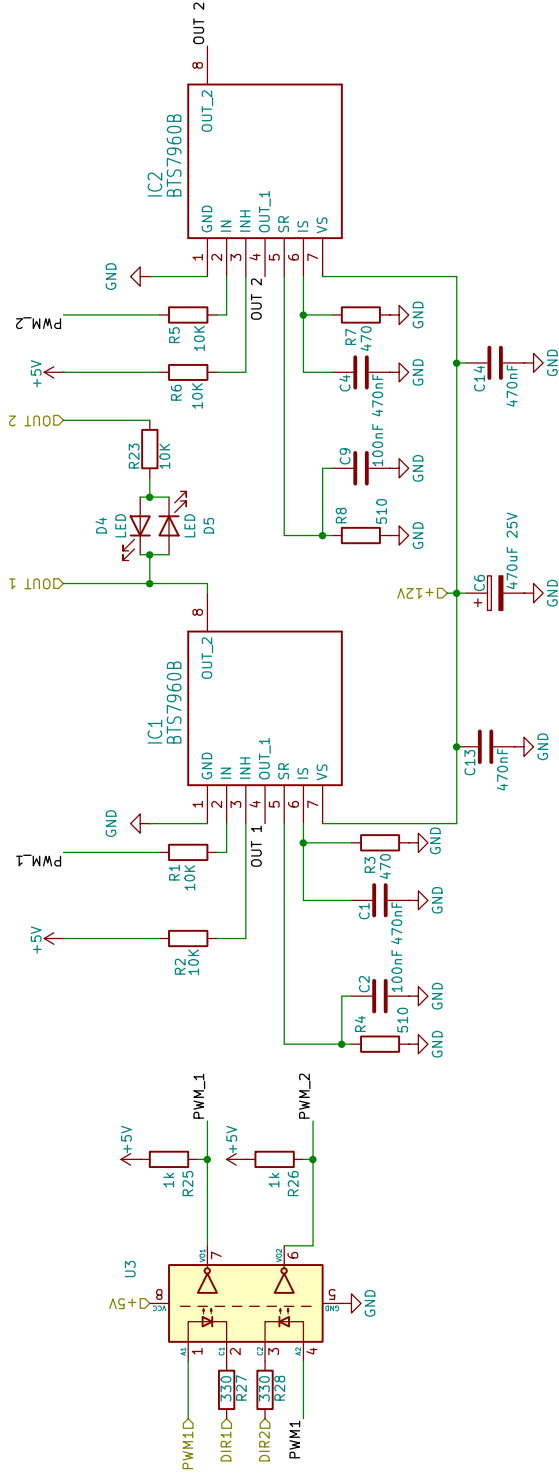


MOTOR 1



RVA ROBOTICS
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Sheet: /MOTOR 1/
File: Motor 1.kicad_sch

Title: Board Robot SOCCER & SUMO

Size: A4 Date: 2026-09-28
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MOTOR 2

The diagram illustrates a circuit for controlling two DC motors. It features two BTS7960B motor drivers (IC3 and IC4) and a 5V voltage regulator (U5).

5V Voltage Regulator (U5): A 5V regulator is connected to a 5V input. Its output is connected to a 1k resistor (R31) and a 1k resistor (R32). The output of R31 is connected to the IN pin of IC3. The output of R32 is connected to the IN pin of IC4.

Motor Driver IC3 (BTS7960B): The IN pin is connected to the output of R31. The SR pin is connected to GND. The VS pin is connected to the +5V input. The OUT_1 pin is connected to the motor terminal 1. The OUT_2 pin is connected to the motor terminal 2.

Motor Driver IC4 (BTS7960B): The IN pin is connected to the output of R32. The SR pin is connected to GND. The VS pin is connected to the +5V input. The OUT_1 pin is connected to the motor terminal 3. The OUT_2 pin is connected to the motor terminal 4.

LED Indicators: Two LEDs (D6 and D7) are connected to the OUT_1 pins of IC3 and IC4, respectively. They are also connected to a 10k resistor (R24) and a 10k resistor (R25).

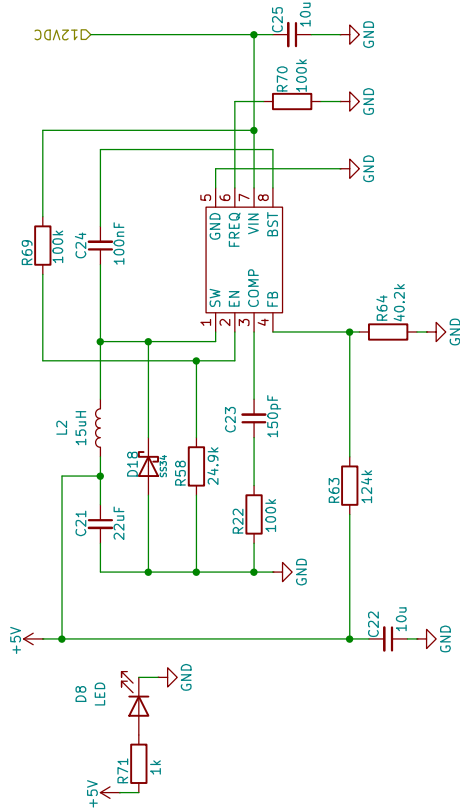
Resistors: 1k resistors (R31, R32) are used for the 5V regulator output. 10k resistors (R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25) are used for various signal conditioning and current limiting.

Capacitors: 100nF capacitors (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C48, C49, C50, C51, C52, C53, C54, C55, C56, C57, C58, C59, C60, C61, C62, C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C86, C87, C88, C89, C90, C91, C92, C93, C94, C95, C96, C97, C98, C99, C100) are used for decoupling and filtering.

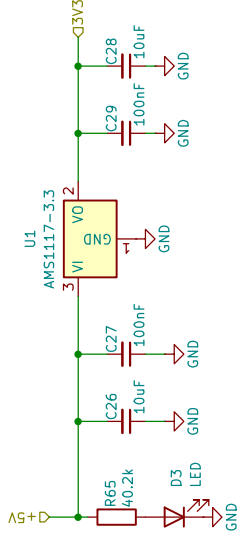
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 File: Motor 2.kicad_sch
Title: Board Robot SOCCER & SUMO
 Size: A4
 Date: 2026-09-28
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12 VDC TO 5 VDC



5 VDC TO 3.3 VDC



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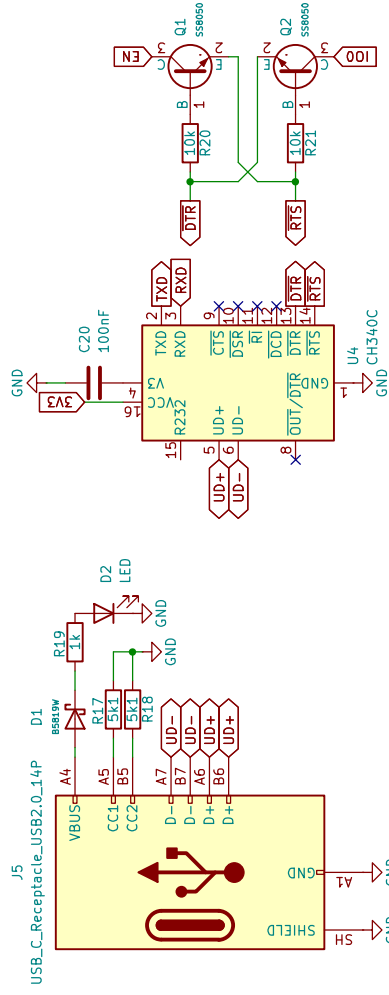
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File: regulator.kicad_sch

Title: Board Robot SOCCER & SUMO

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ESP32 WROOM 32U



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